2 Two atoms X and Y, have masses 3*m* and 2*m* respectively. The 2 atoms move head-on towards each other with the same speed *v* as shown in Fig. 2.1.

Fig. 2.1

Fig 2.2 comprises two velocity-time graphs A and B, which show how the velocity of each atom varies. The interaction between the atoms is elastic.

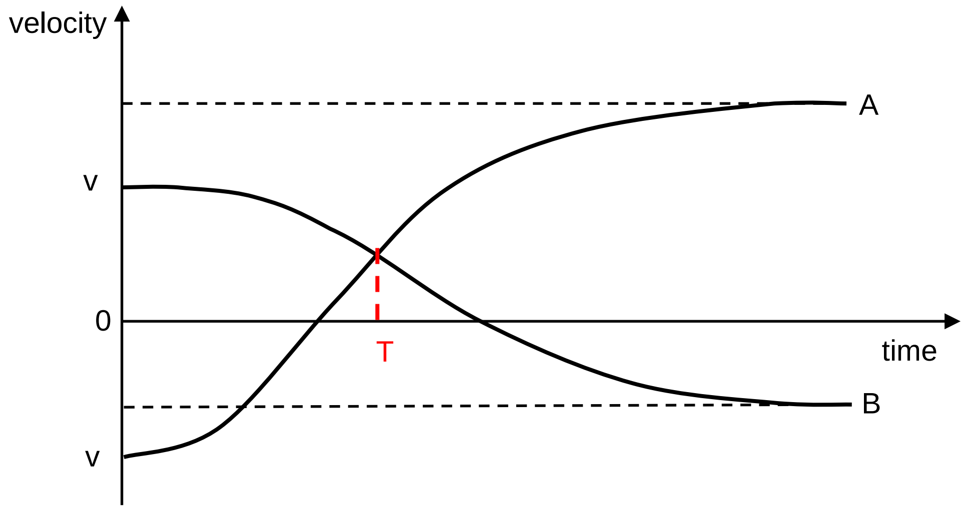

Fig. 2.2 (not to scale)

(a) **(i)** Explain why it is not possible for the atoms to stop at the same instant.

Since the total momentum before the collision is *mv*, it is not possible for both atoms to stop at the same instant as then the total momentum of the system at that instant will be zero. This will violate the principle of conservation of momentum. [1]

(ii) At one instant during the interaction between the atoms, they are both traveling in the same direction with the same speed. Calculate this speed, in terms of *v*.

From the principle of Conservation of momentum:

Total initial momentum = total final momentum

$$3mv - 2mv = 3mu + 2mu$$
$$u = 0.2mv$$

speed = ……………………… [2]

(b) (i) State and explain, which of the curves A or B is the velocity-time sketch for atom Y.

Curve A is the velocity-time sketch for atom Y.

During the collision, by Newton's third law, X and Y will experience a force of equal magnitude and in opposite direction. Since the force on X and Y are the same, and the mass of Y is smaller, Y will experience a larger acceleration and the change in velocity for Y will be larger, which is curve A. [3]

For momentum to be conserved, the magnitude of the change in momentum of X is equal to the magnitude of change in momentum of Y. Since the mass of Y is smaller, the change in velocity for Y will be larger, which is the curve A.

(ii) On Fig. 2.2, mark the instant in time at which the atoms are at their distance of closest approach. Label this point **T**. [1]

(iii) Determine the final speed of each atom in terms of v.

Let V_x be the final speed of X and V_y be the final speed of Y.
Since the collision is elastic:
relative speed of approach = relative speed of separation,

$$v-(-v)=V_Y-V_X$$

$$2v=V_Y-V_X \quad .....(1)$$

By the principle of Conservation of momentum:

$$3mv-2mv=3mV_X+2mV_Y$$

$$v=3V_X+2V_Y \quad ......(2)$$

Solving equation (1) and (2):

$$V_X=-\ 0.6v$$

$$V_Y=1.4v$$

final speed of X = ……………………..

final speed of Y = …………………….. [3]

[Total: 10]

3 A 2.0 kg block on a track is released at A, 1.0 m above the ground as shown in Fig. 3.1. The